Customer Journey Map

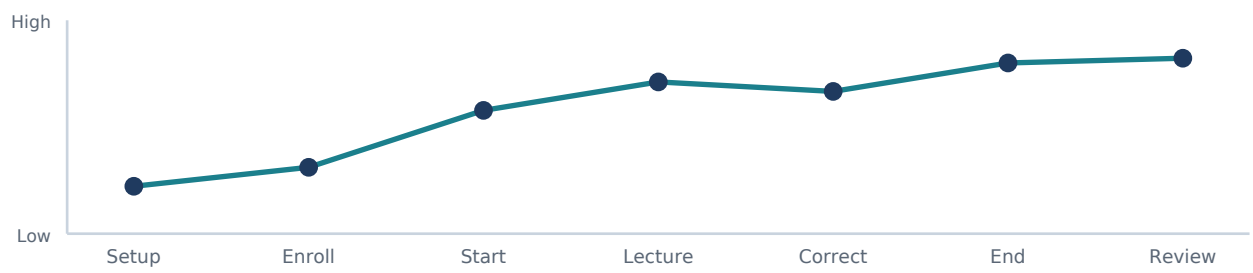
Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

This map follows the **lecturer** through one complete use of the system, from setup to reviewing results. It shows what the user does, what they feel, the pain points of today's manual process, and how the agent responds.

1. Journey Stages

Stage	User actions	Touchpoint	Emotion	Pain point (today)	How the agent helps
1. Setup	Starts backend and dashboard on laptop	Terminal, browser (local host:8501)	Neutral	Tools need internet or special hardware	Two commands; fully offline; status bar shows backend and camera state
2. Enrollment	Registers students with consent and a photo	Enrollment form, Data & Exports page	Cautious	Fear of storing faces without consent	Consent flag is mandatory; no consent means no biometric data saved
3. Start lecture	Enters title and course code, presses Start Session	Live Monitoring page	Confident	Roll call eats lecture time	Attendance starts automatically; no roll call needed
4. During lecture	Teaches while watching the live feed and teaching tips	Live video, Engagement Analytics, Recommendations	Reassured	Cannot tell when attention drops	Live engagement index, tier counts, disengagement alert, tips with action steps
5. Corrections	Fixes an excused or misjudged student	Attendance Management page	Relieved	Errors cannot be fixed or audited	Manual override with a stored reason
6. End lecture	Presses Finalize & End Session	Live Monitoring page	Satisfied	Paper sheets must be typed in later	Final PRESENT / PARTIAL / ABSENT verdicts saved to SQLite
7. Review and export	Opens past sessions, downloads CSV	Student Session Reports, Data & Exports	Satisfied	No history or easy report	Session archive, roster, mean engagement, CSV export

2. Emotional Curve



Illustrative curve based on the pain points and responses in the table above.

3. Opportunities Identified

- Keep setup to two commands so the first impression is positive.
- Show consent and privacy messaging at enrollment to reduce the cautious stage.
- Debounce recommendations (30 s) so the lecture stage stays helpful rather than noisy.
- Provide an audit reason on every manual correction to build trust in the records.